2D Matrices: Factorisation of Determinants

Questions 1–20

Question 1. It is given that $\mathbf{A} = \begin{pmatrix} 4 & 40 \\ 12 & 600 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Find $10\begin{vmatrix} 4 & 4 \\ 12 & 60 \end{vmatrix}$

c) Find $40\begin{vmatrix} 1 & 1 \\ 12 & 60 \end{vmatrix}$

d) Find $480\begin{vmatrix} 1 & 1 \\ 1 & 5 \end{vmatrix}$

Question 2. It is given that $\mathbf{A} = \begin{pmatrix} 120 & 21 \\ 150 & 35 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Find $2\begin{vmatrix} 60 & 21 \\ 75 & 35 \end{vmatrix}$

c) Find $6\begin{vmatrix} 20 & 21 \\ 25 & 35 \end{vmatrix}$

d) Find $30\begin{vmatrix} 4 & 21 \\ 5 & 35 \end{vmatrix}$

e) Find $150\begin{vmatrix} 4 & 21 \\ 1 & 7 \end{vmatrix}$

f) Find $1050\begin{vmatrix} 4 & 3 \\ 1 & 1 \end{vmatrix}$

g) Hence write 1050 as a product of its prime factors.

Question 3. It is given that $\mathbf{A} = \begin{pmatrix} k & k+1 \\ 2k & 4(k+1) \end{pmatrix}$ where k is a constant.

a) Find $|\mathbf{A}|$

b) Find $k\begin{vmatrix} 1 & k+1 \\ 2 & 4(k+1) \end{vmatrix}$

c) Find $2k\begin{vmatrix} 1 & k+1 \\ 1 & 2(k+1) \end{vmatrix}$

d) Find $2k(k+1)\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix}$

e) Hence write $|\mathbf{A}|$ in fully factorised form.

Question 4. It is given that $\mathbf{A} = \begin{pmatrix} 18k-9 & 3-k \\ 6k-3 & 2k-6 \end{pmatrix}$ where k is a constant.

a) Find $|\mathbf{A}|$

b) Find $(2k-1)\begin{vmatrix} 9 & 3-k \\ 3 & 2k-6 \end{vmatrix}$

c) Find $3(2k-1)\begin{vmatrix} 3 & 3-k \\ 1 & 2k-6 \end{vmatrix}$

d) Find $3(2k-1)(3-k)\begin{vmatrix} 3 & 1 \\ 1 & -2 \end{vmatrix}$

e) Hence write $|\mathbf{A}|$ in fully factorised form.

Question 5. It is given that $\mathbf{A} = \begin{pmatrix} 4k & 6k-2 \\ 20k & 9k-3 \end{pmatrix}$, where k is a constant. Completely factorise $|\mathbf{A}|$ without evaluating a determinant unless it is completely independent of k

Question 6. It is given that $\mathbf{A} = \begin{pmatrix} 4-8k & 5-10k \\ 3k-6 & 4k-8 \end{pmatrix}$ where k is a constant. By factorising $|\mathbf{A}|$, solve the equation $|\mathbf{A}| = 0$

Question 7. It is given that $\mathbf{A} = \begin{pmatrix} 10 & 8 \\ 8 & 8 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Factor 2 out of R1
- Factor 8 out of R2
- Let the new R1 = R1 − 4R2

c) Use your answer to b) to find $|\mathbf{A}|$

Question 8. It is given that $\mathbf{A} = \begin{pmatrix} 16 & 20 \\ 9 & 5 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Factor 5 out of C2
- Let the new C1 = C1 − 4C2

c) Use your answer to b) to find $|\mathbf{A}|$

Question 9. It is given that $\mathbf{A} = \begin{pmatrix} 40 & 35 \\ 55 & 25 \end{pmatrix}$

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively, and C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new C1 = C1 − 2C2
- Let the new R1 = R1 + 6R2

c) Use your answer to b) to find $|\mathbf{A}|$

Question 10. It is given that $\mathbf{A} = \begin{pmatrix} 4k+5 & 53 \\ k & 12 \end{pmatrix}$ where k is a constant.

a) Find $|\mathbf{A}|$

b) Let R1 and R2 be the first and second rows of $|\mathbf{A}|$ respectively, and C1 and C2 be the first and second columns of $|\mathbf{A}|$ respectively. Perform the following steps in order:

- Let the new R1 = R1 − 4R2
- Let the new C2 = C2 − C1

c) Use your answer to b) to find $|\mathbf{A}|$

Question 11. Grace is attempting to factorise the determinant of the matrix $\mathbf{A} = \begin{pmatrix} q+1 & -5p \\ 3q+3 & 15p \end{pmatrix}$ where p and q are constants.
Her working and annotations are shown below:

	$\begin{vmatrix} q+1 & -5p \\ 3q+3 & 15p \end{vmatrix}$
(1)	$= \begin{vmatrix} q+1 & -5p \\ 3(q+1) & 15p \end{vmatrix}$
(2) Factor $(q+1)$ out of C1	$= (q+1)\begin{vmatrix} 1 & -5p \\ 3 & 15p \end{vmatrix}$
(3) Factor 3 out of C1	$= (q+1)\begin{vmatrix} 1 & -5p \\ 1 & 15p \end{vmatrix}$
(4) Factor $5p$ out of C2	$= 5p(q+1)\begin{vmatrix} 1 & -1 \\ 1 & 3 \end{vmatrix}$
(5) Let the new C1 = C1 + C2:	$= 5p(q+1)\begin{vmatrix} 0 & -1 \\ 4 & 3 \end{vmatrix}$
(6) Then:	$= 20p(q+1)$

a) Identify where Grace has made a mistake and explain what she has done wrong.
b) Write out fully correct workings for this question.

Question 12. Fully factorise each of these determinants. Your aim in each case before evaluating the determinant is to have at least one element being 0

a)
$$\begin{vmatrix} x & y \\ x^2 & y^2 \end{vmatrix}$$

b)
$$\begin{vmatrix} x^2 & y^2 \\ xy & xy \end{vmatrix}$$

c)
$$\begin{vmatrix} xy & xy^2 \\ x^2 & xy \end{vmatrix}$$

d)
$$\begin{vmatrix} yz & xy \\ xyz & xyz \end{vmatrix}$$

e)
$$\begin{vmatrix} xyz & yz \\ xy & xyz \end{vmatrix}$$

f)
$$\begin{vmatrix} x^2y & xz^2 \\ xy^2 & xyz^2 \end{vmatrix}$$

g)
$$\begin{vmatrix} x^2y^2z & xy^2z^2 \\ x^2yz^2 & xyz \end{vmatrix}$$

Question 13. Fully factorise each of these determinants, where p and q are constants. Your aim in each case before evaluating the determinant is to have at least one element being 0

a)

$$\begin{vmatrix} p & -p-2q \\ 4p+5q & p+2q \end{vmatrix}$$

b)

$$\begin{vmatrix} 2q-2p & 3p-3q \\ 14q-6p & 2p \end{vmatrix}$$

c)

$$\begin{vmatrix} 16p+5q-13 & 4-8p \\ 6q-12p & 6p-3 \end{vmatrix}$$

d)

$$\begin{vmatrix} 3-2q & 4q-8p \\ 6-4q & 8q+24p-30 \end{vmatrix}$$

Question 14. Given that

$$|\mathbf{A}| = xy \begin{vmatrix} 1 & -3 \\ 2 & 4 \end{vmatrix}$$

show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} x & -3y \\ 0 & 10y \end{vmatrix}$$

Question 15. Given that

$$|\mathbf{A}| = x(x-1)y\begin{vmatrix} 2 & -3 \\ 3 & -1 \end{vmatrix}$$

show clearly, by using row and column operations, that

$$|\mathbf{A}| = \begin{vmatrix} 0 & 7y - 7xy \\ x & x + 2y - 2xy \end{vmatrix}$$

Question 16. It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} x - 5a & -4a \\ 14x + 2a & 9x + 7a \end{vmatrix}$$

where a is a constant. Find the possible values of x in terms of a

Question 17. It is given that $\mathbf{A}$ is a singular matrix such that

$$\det \mathbf{A} = \begin{vmatrix} 8a^2 - 2x^2 & 3x^2 - 6ax - 24a^2 \\ 5ax - 10a^2 & 8ax - 32a^2 \end{vmatrix}$$

where a is a constant such that $a \neq 0$.
Find the possible values of x in terms of a

Question 18. For constants a, b and c, fully factorise

$$\begin{vmatrix} 3a - 4b - 3c & a - 2b - 2c \\ -a + 4b - c & a + 2b - 2c \end{vmatrix}$$

Question 19. For constants p and q, fully factorise

$$\begin{vmatrix} pq - 4p + 5q - 20 & 3pq + 15q \\ 3pq - 12p - 6q + 24 & 4pq - 10p - 8q + 20 \end{vmatrix}$$

Question 20.

a) For constants a and b, fully factorise

$$\begin{vmatrix} x + 3a + b & x + 3a \\ 5a + b & 5a \end{vmatrix}$$

b) It is given that the transformation T can be represented by the matrix

$$A = \begin{pmatrix} x + 35 & x + 30 \\ 55 & 50 \end{pmatrix}$$

When a 2D shape of area 20 cm^2 is transformed by T, its new area is 50 cm^2.
Use your answer to part a) to find the possible values of x